

es on collecting low-value flexible waste to turn into shrink and stretch wraps. She demonstrates a plastic wrap that is used by The House of Anita Dongre, a fashion apparel brand. “It is 100 per cent recyclable and made from 60 per cent post-consumer recycled plastic material,” she says.

In October 2022, the Government of India introduced a law requiring anyone manufacturing or using plastic to introduce a recycled version in their packaging. “That was a huge tipping point for the industry,” says Dewett.

Dealing with waste in every sense is a “dirty business,” but it is a job that must be done. Take the case of Ahmedabad-based Ishitva Robotic Systems that makes AI/ML and IoT-based automation solutions to sort recyclable materials. “Technology is our competitive edge and we are the enabler to create an ecosystem. We are the tech providers and this is a very scalable model,” says Jitesh Dadlani, Founder and CTO of the start-up. He cites the Swachh Bharat Mission as a big catalyst for his business. “We saw the opportunity in the value chain and it became possible to build solutions based in Industry 4.0 tools. Today, effective sorting is still the weakest link in the process of creating a circular value chain,” he says.

In a nascent industry, there is no limit to innovation, and looking for viable alternatives is one such example. Neha Jain, Founder & CEO of Zero Circle—that uses seaweed to create bio-alternatives that can replace plastics—has gone for the “top-down” model to keep plastics out of the picture. After travelling frequently to coastal Maharashtra, she hit upon the idea of using seaweed (that has 90 per cent water), as the raw material for her



“It was apparent that plastic was ripe for an intervention, since there were major innovations emerging in the space, with a clear need for an ecosystem to be built around it”

HARSH MARIWALA
CHAIRMAN, MARICO;
FOUNDER, MARICO
INNOVATION
FOUNDATION



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FOUNDER AND CTO,
ISHITVA ROBOTIC SYSTEMS

bioplastics “because of its higher biomass, fast reproduction and cost-effectiveness.” Jain says she works with the manufacturers who make the final packaging that go out to the users. “There is plastic hidden even in food containers made of paper. Seaweed-based coatings are a viable alternative to the plastics used in them and are high-performing.” With her product nearing the end of research and development, she is preparing to go into production shortly and is optimistic about what lies ahead.

Of course, the process of moving to alternatives would be a challenge, says Mariwala. He thinks that since the alternatives are expensive compared to virgin plastic, the government should provide incentives for the production of alternatives and benefits such as less taxes or tax breaks over the short- to medium term. This will help the alternatives achieve scale, which will bring down their prices, further fueling their adoption.

By any yardstick, this transition will take time, but it is one that is non-negotiable. For over 150 years, the world has used plastic and the moment for a complete relook at how plastics are used is upon us, says Mariwala. The circular economy is a good starting point. With start-ups innovating new products and processes in the space, there is a way to create a robust and profitable business model. However, it needs substantial assistance from the government and other stakeholders at every stage. For the sake of a better environment and to ensure the generations that follow us live well, the plastic story needs to be addressed—smartly and quickly. **BT**

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